

Repair notes for `hyper-lean`

A global positive content, its uniqueness, reciprocity,
and a corrected transfer principle

Draft for the authors of `algebraic-probability.tex`

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Abstract

The probability manuscript leaves one central item open: a single geometric content on a specified event algebra that is finitely additive, independent of decomposition and strictly positive. We show that this follows from classical valuation theory (Groemer’s extension theorem and Hadwiger’s characterization) together with one short new argument, a leading-coefficient positivity lemma in the ordered field $\mathbb{R}(\varepsilon)$. We also prove a uniqueness statement from cube calibration, an Ehrhart–Macdonald-type reciprocity for relatively open polytopes, an elementary self-contained version for the box algebra suitable for immediate Lean formalization, and we record how to repair the transfer axiom of the older introductory manuscript. Classical inputs are cited, not reproved; everything else is proved here and should be checked.

1 Setting and classical inputs

Throughout, $d \geq 1$, $\mathbb{H} = \mathbb{R}(\varepsilon)$ is ordered with $0 < \varepsilon < r$ for all real $r > 0$, and every geometric datum is *standard*: a subset of $\mathbb{R}^d$ defined from real numbers. Fix a compact convex *board* $\Omega \subseteq \mathbb{R}^d$ with nonempty interior. Let $\mathcal{K}(\Omega)$ be the nonempty compact convex subsets of Ω , and let $\mathcal{A}(\Omega)$ be the Boolean algebra of subsets of Ω generated by $\mathcal{K}(\Omega)$ (complements taken in Ω). Elements of $\mathcal{A}(\Omega)$ are our events. It contains points, segments, closed disks, closed balls, polytopes, all their finite unions, and all set differences such as half-open segments, a disk minus a diameter, or relatively open polytopes.

Sign rule in $\mathbb{H}$. A nonzero element $x = \sum_{i \geq i_0} a_i \varepsilon^i$ (finite sum, $a_{i_0} \neq 0$) has the sign of a_{i_0} . All contents below are of this polynomial form.

Write $V_0, \dots, V_d$ for the intrinsic volumes, normalized so that $V_k(K)$ is the k -dimensional volume $\mathcal{H}^k(K)$ when $\dim K = k$, and $V_j(K) = 0$ for $j > \dim K$. V_0 is the Euler characteristic on convex sets ($V_0 \equiv 1$ on $\mathcal{K}$).

- (G) *Groemer’s extension theorem* [1, Ch. 6], [2]. Each V_j extends uniquely to a linear functional φ_j on the real vector space $S(\Omega)$ spanned by the indicators $\mathbf{1}_K$, $K \in \mathcal{K}(\Omega)$; that is, $\varphi_j(\mathbf{1}_K) = V_j(K)$.
- (H) *Hadwiger’s theorem* [3, 2]. Every real-valued valuation on $\mathcal{K}(\mathbb{R}^d)$ that is continuous in the Hausdorff metric and invariant under rigid motions is a real linear combination of $V_0, \dots, V_d$.
- (M) *McMullen’s Euler-type relation* [4]. If φ is a translation-invariant valuation on polytopes, homogeneous of degree j , then for every polytope P , $\sum_F (-1)^{\dim F} \varphi(F) = (-1)^j \varphi(-P)$, the sum over the nonempty faces F of P including P .

2 The global content

Lemma 2.1 (Indicators). *For every $E \in \mathcal{A}(\Omega)$, $\mathbf{1}_E \in S(\Omega)$. More precisely, if E lies in the algebra generated by $K_1, \dots, K_m \in \mathcal{K}(\Omega)$, then E is a disjoint union of nonempty atoms*

$$A_T = \left(\Omega \cap \bigcap_{i \in T} K_i \right) \setminus \bigcup_{i \notin T} K_i, \quad T \subseteq \{1, \dots, m\},$$

and each atom has the form $C \setminus F$, with C compact convex and F a finite union of compact convex sets.

Proof. The atoms partition Ω , and every set in the generated algebra is a union of atoms. Further, $\mathbf{1}_{A_T} = \prod_{i \in T} \mathbf{1}_{K_i} \cdot \mathbf{1}_\Omega \prod_{i \notin T} (\mathbf{1}_\Omega - \mathbf{1}_{K_i})$; expanding gives a signed sum of indicators of finite intersections, which are compact convex or empty. Finally $C = \Omega \cap \bigcap_T K_i$ and $F = \bigcup_{i \notin T} K_i$. $\square$

Definition 2.2 (Global content). For $E \in \mathcal{A}(\Omega)$ put

$$\nu_\varepsilon(E) = \sum_{j=0}^d \varphi_j(\mathbf{1}_E) \varepsilon^{d-j} \in \mathbb{H}.$$

For $K \in \mathcal{K}(\Omega)$ this is $\sum_j V_j(K) \varepsilon^{d-j} = \varepsilon^d g_K(\varepsilon^{-1})$, the rescaled Wills polynomial used in the manuscript.

Lemma 2.3 (Low-dimensional unions). *Let $F = K_1 \cup \dots \cup K_m$ with K_i compact convex and $\dim K_i \leq k$. Then $\varphi_j(\mathbf{1}_F) = 0$ for $j > k$ and $\varphi_k(\mathbf{1}_F) = \mathcal{H}^k(F)$.*

Proof. Inclusion–exclusion gives $\mathbf{1}_F = \sum_{\emptyset \neq U} (-1)^{|U|+1} \mathbf{1}_{K_U}$ with $K_U = \bigcap_{i \in U} K_i$, convex of dimension $\leq k$. Apply φ_j and the normalization of V_j : terms with $j > k$ vanish, and for $j = k$ we obtain $\sum (-1)^{|U|+1} \mathcal{H}^k(K_U)$, where $\mathcal{H}^k(K_U) = V_k(K_U)$ holds also when $\dim K_U < k$ (both sides are 0). Since $\mathcal{H}^k$ is a measure, the same inclusion–exclusion sum equals $\mathcal{H}^k(F)$. $\square$

Lemma 2.4 (Relatively open pieces are fat). *Let C be compact convex of dimension k and let $U \subseteq C$ be nonempty and open in C . Then $\mathcal{H}^k(U) > 0$.*

Proof. Pick $x \in U$ and $r > 0$ with $C \cap B(x, r) \subseteq U$. For small $t > 0$, the homothetic copy $x + t(C - x)$ lies in C (convexity) and in $B(x, r)$. Hence $\mathcal{H}^k(U) \geq t^k \mathcal{H}^k(C) > 0$. $\square$

Theorem 2.5 (Global positive content). *The map $\nu_\varepsilon : \mathcal{A}(\Omega) \rightarrow \mathbb{H}$ is well defined and*

1. *finitely additive: $\nu_\varepsilon(E \cup E') = \nu_\varepsilon(E) + \nu_\varepsilon(E')$ for disjoint E, E' ; in particular it is independent of any decomposition;*
2. *equal to $\varepsilon^d g_K(\varepsilon^{-1})$ on compact convex K ;*
3. *invariant under rigid motions g with $gE \subseteq \Omega$;*
4. *strictly positive: $\nu_\varepsilon(E) > 0$ for every nonempty E ;*
5. *of known leading order: if k is the largest dimension of a convex set C occurring in a nonempty atom $C \setminus F$ of E , then $\nu_\varepsilon(E) = \mathcal{H}^k(E) \varepsilon^{d-k} + O(\varepsilon^{d-k+1})$ with $\mathcal{H}^k(E) > 0$.*

Proof. (1) The definition uses only $\mathbf{1}_E$, and $\mathbf{1}_{E \cup E'} = \mathbf{1}_E + \mathbf{1}_{E'}$ for disjoint sets; apply linearity of each φ_j (Lemma 2.1 and (G)). (2) is the definition of φ_j on $\mathcal{K}$. (3) The functional $\mathbf{1}_E \mapsto \varphi_j(\mathbf{1}_{gE})$ extends V_j , which is motion invariant; uniqueness in (G) gives invariance.

(4) and (5). By (1) and Lemma 2.1 it suffices to treat one nonempty atom $A = C \setminus F$ with $\dim C = k$; a sum of positive elements is positive and the leading orders add as stated. Write $\nu_\varepsilon(A) = \nu_\varepsilon(C) - \nu_\varepsilon(C \cap F)$. The set $C \cap F$ is a finite union of compact convex sets of dimension $\leq k$, so Lemma 2.3 and the normalization of V_j on C show that all coefficients of ε^{d-j} with $j > k$ vanish, while the coefficient of ε^{d-k} is

$$\mathcal{H}^k(C) - \mathcal{H}^k(C \cap F) = \mathcal{H}^k(C \setminus F) > 0$$

by Lemma 2.4, since $C \setminus F$ is nonempty and open in C . The remaining terms carry higher powers of ε with real coefficients, so the sign rule gives $\nu_\varepsilon(A) > 0$. $\square$

Corollary 2.6 (Regular finitely additive probability). *$P(E) = \nu_\varepsilon(E)/\nu_\varepsilon(\Omega)$ is a finitely additive probability on $\mathcal{A}(\Omega)$ with $P(E) > 0$ for every nonempty event. Hence $P(\cdot | E)$ is defined for every nonempty E , and every finite context of the manuscript (square, cube, disk and ball darts) is the restriction of this one probability to a finite subalgebra. Moreover $\text{st } P(E) = \lambda(E)/\lambda(\Omega)$, so classical geometric probability is the standard part.*

Proof. Positivity and additivity are Theorem 2.5; $\nu_\varepsilon(\Omega)$ has leading term $\lambda(\Omega) > 0$ at ε^0 . For the standard part, divide the two leading coefficients at order ε^0 ; these are $\varphi_d(\mathbf{1}_E) = \lambda(E)$ (Lebesgue measure is the additive extension of V_d) and $\lambda(\Omega)$. $\square$

Remark 2.7 (Where standardness is used). Only in Lemma 2.4 and the sign rule: coefficients of $\nu_\varepsilon(E)$ must be real. The manuscript's example $(\varepsilon/2)\varepsilon - \varepsilon^2 < 0$ uses an infinitesimal length, i.e. a nonstandard coefficient; this is exactly what the theorem excludes. Countable additivity is not available: in dimension one, $(0, 1] = \bigsqcup_n (\frac{1}{n+1}, \frac{1}{n}]$, each half-open piece has content equal to its length (the endpoint terms cancel), and the partial sums $1 - \frac{1}{N+1}$ never come within ε of $\nu_\varepsilon((0, 1]) = 1$, so they do not converge in the order topology of $\mathbb{H}$.

3 Uniqueness from cube calibration

Theorem 3.1 (Uniqueness). *Let $W \subseteq \mathbb{H}$ be a finite-dimensional real subspace and $\mu : \mathcal{K}(\mathbb{R}^d) \rightarrow W$ a valuation that is Hausdorff continuous and invariant under rigid motions. If $\mu([0, a]^d) = (a + \varepsilon)^d$ for $d + 1$ distinct values $a > 0$, then $\mu = \varepsilon^d g_K(\varepsilon^{-1})$ on $\mathcal{K}(\mathbb{R}^d)$, and μ has exactly one additive extension to $\mathcal{A}(\Omega)$, namely ν_ε .*

Proof. Apply (H) to each coordinate of μ with respect to a real basis of W : $\mu = \sum_j c_j V_j$ with $c_j \in W$. Since $V_j([0, a]^d) = \binom{d}{j} a^j$, the hypothesis reads $\sum_j \binom{d}{j} c_j a^j = \sum_j \binom{d}{j} \varepsilon^{d-j} a^j$ at $d + 1$ points, so the two polynomials of degree $\leq d$ in a coincide and $c_j = \varepsilon^{d-j}$. An additive extension is determined on atoms by Lemma 2.1, because $\mathbf{1}_{A_T}$ is an integer combination of indicators of convex sets. $\square$

Remark 3.2. The manuscript's own calibration, $\nu_\varepsilon([0, a]) = a + \varepsilon$ in dimension one plus the product rule on boxes, implies the hypothesis. Continuity and motion invariance are genuine assumptions; without continuity, Hadwiger's theorem fails.

4 Reciprocity for relatively open polytopes

Theorem 4.1 (Infinitesimal Ehrhart–Macdonald reciprocity). *Let $P \subseteq \Omega$ be a polytope of dimension k . Then*

$$\nu_\varepsilon(\text{relint } P) = \sum_{j=0}^k (-1)^{k-j} V_j(P) \varepsilon^{d-j} = (-1)^k \varepsilon^d g_P(-\varepsilon^{-1}).$$

Proof. The relative interiors of the nonempty faces partition P . Möbius inversion on the face lattice (an Eulerian poset) gives $\mathbf{1}_{\text{relint } P} = \sum_F (-1)^{k-\dim F} \mathbf{1}_F$. Apply φ_j and (M) with $\varphi = V_j$, using $V_j(-P) = V_j(P)$: $\varphi_j(\mathbf{1}_{\text{relint } P}) = (-1)^k (-1)^j V_j(P)$. $\square$

Checks: an open segment of length L gives $L\varepsilon^{d-1} - \varepsilon^d$, as in the manuscript; an open square of side a in the plane gives $a^2 - 2a\varepsilon + \varepsilon^2 = (a - \varepsilon)^2$; an open cube gives $(a - \varepsilon)^3$. In general a relatively open box is the product $\prod(a_i - \varepsilon)$, the mirror image of $\prod(a_i + \varepsilon)$ for the closed box.

5 The box algebra: an elementary version for Lean

The following is self-contained, needs no valuation theory, and is the recommended first formalization target.

Let $\mathcal{B}_d$ be the algebra of subsets of $[0, 1]^d$ generated by closed axis-parallel boxes with real corners. In dimension one define, for $E \in \mathcal{B}_1$,

$$\nu_1(E) = \lambda(E) + \varepsilon \chi(E), \quad \chi(E) = \sum_{x \in [0, 1]} (\mathbf{1}_E(x) - \mathbf{1}_E(x^+)),$$

where $\mathbf{1}_E(x^+)$ is the right limit (taken as 0 at $x = 1$); the sum is finite because $\mathbf{1}_E$ is a step function.

Proposition 5.1 (One dimension). *ν_1 is finitely additive on $\mathcal{B}_1$, $\nu_1([a, b]) = b - a + \varepsilon$, $\nu_1(\{a\}) = \varepsilon$, $\nu_1((a, b)) = b - a - \varepsilon$, and $\nu_1(E) > 0$ for every nonempty E .*

Proof. Both λ and χ are linear in $\mathbf{1}_E$, which gives additivity with no choice of decomposition. The three values follow by evaluating the jump sum. If $\lambda(E) > 0$ the content is positive by the sign rule; otherwise E is a finite nonempty set of points and $\chi(E) = \#E > 0$. $\square$

Proposition 5.2 (Boxes by Fubini). *Define ν_d on $\mathcal{B}_d$ inductively by integrating $x_1 \mapsto \nu_{d-1}(E_{x_1})$, a step function, against ν_1 , where $\int f d\nu_1 = \int f d\lambda + \varepsilon \sum_x (f(x) - f(x^+))$. Then ν_d is finitely additive, satisfies $\nu_d(\prod I_i) = \prod \nu_1(I_i)$, is strictly positive on nonempty sets, and coincides with ν_ε on $\mathcal{B}_d$.*

Proof sketch. Additivity is linearity of the integral in the integrand. Products factor by construction. For positivity, every nonempty E is a disjoint union of nonempty grid cells $\prod I_i$ with each I_i a point or an open interval; each cell has content $\prod \nu_1(I_i) > 0$. Agreement with ν_ε follows from Theorem 3.1 on boxes, or directly since both are additive and agree on closed boxes. $\square$

Lean status. Proposition 5.1 is checked in `Hyper/IntervalAlgebra.lean`, for events presented as finite lists of pairwise disjoint points and open intervals (every element of $\mathcal{B}_1$ has such a presentation; that representation lemma is not formalized). The file proves `content_eq_of_set_eq` (two presentations of the same set have the same content), `content_pos`, and the calibration `closed_content`; the proof is exactly the representation-free formula $\nu_1 = \lambda + \varepsilon \chi$ above, with `volume` from Mathlib and χ as a jump sum. The axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`. Next step: Proposition 5.2 by induction on d , reusing the products in `CubicContent.lean`.

6 Repairing the transfer axiom of the older manuscript

The older text asserts a transfer principle for $\mathbb{R}(\varepsilon)$. This is false: the sentence $\forall x (x > 0 \rightarrow \exists y y \cdot y = x)$ holds in $\mathbb{R}$ but fails in $\mathbb{R}(\varepsilon)$ at $x = \varepsilon$. Three repairs exist, increasing in strength.

1. **Real closure (semialgebraic transfer).** Let $\mathbb{R}(\varepsilon)^{\text{rc}}$ be the real closure, realized inside algebraic Puiseux series $\sum_{q \in \frac{1}{n}\mathbb{Z}} a_q \varepsilon^q$. Real closed fields are model complete (Tarski, A. Robinson) [5], so $\mathbb{R} \preceq \mathbb{R}(\varepsilon)^{\text{rc}}$: every first-order statement in $+, \cdot, <$ with real parameters transfers. This covers all semialgebraic calculus (polynomial and algebraic functions, the intermediate and extreme value theorems for them, and derivatives as standard parts of difference quotients).
2. **Hahn series with restricted analytic functions.** The Hahn field $\mathbb{R}((\varepsilon^{\mathbb{Q}}))$ is real closed and, with restricted analytic functions given by convergent Taylor expansion, is an elementary extension of $\mathbb{R}_{\text{an}}$ [6]. Thus $\sin, \cos, \exp$ on bounded arguments transfer. The repository's rational-exponent HyperList backend embeds in this field, so this route fits the existing code (`Hyper/theory/HyperHahn.lean`).
3. **Exponentiation and full transfer.** Logarithmic-exponential series give an elementary extension of $\mathbb{R}_{\text{an}, \exp}$ [7]. No o-minimal structure can transfer $\sin$ on infinite arguments, since $\sin$ on $\mathbb{R}$ is not definable there; for that, and for transfer of arbitrary internal sets, an ultrapower is required, at the price of a non-canonical ε .

The honest restatement for the introductory manuscript is therefore route 1: $\mathbb{H}^{\text{rc}}$ with semialgebraic transfer, plus route 2 for elementary transcendental functions at finite arguments.

Status summary

Statement	Status
Global additive content on $\mathcal{A}(\Omega)$ (Thm. 2.5(1–3))	classical, via (G)
Strict positivity and leading order (Thm. 2.5(4–5))	proved here; short
Uniqueness from cube calibration (Thm. 3.1)	proved here from (H)
Reciprocity $\nu_\varepsilon(\text{relint } P) = (-1)^k \varepsilon^d g_P(-\varepsilon^{-1})$	proved here from (M); likely known in another guise
Interval algebra (Prop. 5.1)	Lean-checked
Box algebra in dimension d (Prop. 5.2)	elementary; next Lean target
Transfer for $\mathbb{R}(\varepsilon)$	false; replaced by routes 1–3
Full $\mathcal{A}(\Omega)$ in Lean	needs intrinsic volumes, not in Mathlib

References

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